

Design and Implementation of a ~25-Yr Pseudoprospective Earthquake Forecasting Experiment in China (AoyuX)



Jiawei Li¹ , Qingyuan Zhang^{1,2,3,4} , and Didier Sornette^{*1}

Abstract

Forecast models in statistical seismology are commonly evaluated with log-likelihood (LL) scores of the full distribution $P(n)$ of earthquake numbers, yet fat tails and out-of-range observations (realized counts exceeding all simulations) can bias model ranking. We develop a tail-aware evaluation framework that estimates cell-wise $P(n)$ using adaptive Gaussian kernel density estimation and tests three strategies for handling out-of-range counts (power law extrapolation, fixed minimum probability, and exclusion). Using the AoyuX platform, we perform an ~25-yr (2000–2023) month-by-month pseudoprospective forecast experiment in the China Seismic Experimental Site (CSES), comparing epidemic-type aftershock sequence (ETAS) model with a homogeneous background (ETAS _{μ}) with a spatially heterogeneous variant (ETAS _{$\mu(x,y)$}) across six spatial resolutions and five magnitude thresholds; each forecast window is supported by 100,000 simulations. Empirical frequency distributions of counts per cell are well described by power laws $P(n) \sim n^{-(1+a)}$ with exponents $a \approx 1.02 \pm 0.23$ across all settings. Using previous theoretical results, this provides a robust estimate of the productivity exponent, $\alpha = 0.83 \pm 0.19$ using a b value equal to 0.85 ± 0.002 , providing a valuable quantification of this key parameter in aftershock modeling. Model ranking is sensitive to how the tail of the full distribution $P(n)$ of earthquake counts is treated: power law extrapolation is both theoretically justified and empirically the most robust, whereas the apparent superiority of the fixed-probability approach in a few coarse low-magnitude cases can be attributed to noise. Cumulative information gain (CIG) shows that ETAS _{$\mu(x,y)$} outperforms ETAS _{μ} in data-rich configurations (e.g., $M_t = 3.0$ to 3.5 and whole-region tests), with positive mean information gain spatially concentrated in regions where $M \geq 5$ earthquakes occurred during the study period, whereas in data-poor settings, stochastic fluctuations dominate. A coefficient-of-variation analysis of per-window LL differences distinguishes genuine upward trends in CIG from noise-dominated fluctuations. By aligning a fat tail-aware scoring methodology with an open testing platform, our work advances fair and statistically grounded assessment of earthquake forecasting models for the CSES and beyond.

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Introduction

Advancing earthquake forecasting (probabilistic) and/or prediction (deterministic) benefits from a transparent and iterative research paradigm: build models that explain observed seismicity; test them prospectively or pseudoprospectively against withheld or genuinely future data; use the resulting feedback to validate assumptions; and then retain, revise, or replace model components before repeating the cycle. Such a loop of construction, testing, correction, and refinement underpins cumulative progress in the field (Kagan and Jackson, 2000; Werner *et al.*, 2011; Ogata, 2022). Crucial to this paradigm is rigorous, unbiased performance evaluation: determining, with statistical

significance, whether a candidate model truly outperforms a designated benchmark (the null hypothesis H_0) rather than

1. Institute of Risk Analysis, Prediction and Management (Risks-X), Academy for Advanced Interdisciplinary Studies, Southern University of Science and Technology (SUSTech), Shenzhen, China, <https://orcid.org/0000-0003-1226-8162> (JL); <https://orcid.org/0009-0001-8236-200X> (QZ); <https://orcid.org/0000-0001-7874-7962> (DS); 2. School of Systems Science, Beijing Normal University, Beijing, China; 3. International Academic Center of Complex Systems, Beijing Normal University, Zhuhai, China; 4. Department of Systems Science, Faculty of Arts and Sciences, Beijing Normal University, Zhuhai, China

*Corresponding author: dsornette@ethz.ch

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relying on ad hoc or subjective criteria. The international Collaboratory for the Study of Earthquake Predictability (CSEP) and RichterX global earthquake forecasting platform formalized this ethos with standardized tests, scoring rules, and community benchmarks, demonstrating both the value and the challenges of fair model comparison (Schorlemmer *et al.*, 2007, 2010; Taroni *et al.*, 2018; Kamer *et al.*, 2021; Nandan, Kamer, *et al.*, 2021; Bayliss *et al.*, 2022).

In practice, a widely adopted evaluation framework is to run pseudoprospective experiments over consecutive, nonoverlapping time windows. Genuine prospective forecasting, evaluation on data not yet observed, should be stressed, with pseudoprospective experiments serving as a preparatory or complementary step when fully prospective deployment is not yet feasible. For each time window and for each model–benchmark pair, one computes the full probability distribution or probability density function (PDF) $P(n)$ of earthquake numbers n at the grid-cell level (given a chosen magnitude threshold M_i and spatial resolution) and then scores the model by the LL obtained as the logarithm of the products over all cells of the $P(n_i)$ s, in which n_i are the observed counts in each i th grid (see the [Pseudoprospective forecasting experiments](#) section for more details). Forecasting and scoring the distribution, rather than only its mean rate, has been shown to be both fair and robust, and it aligns naturally with aftershock-cascade physics (Ogata *et al.*, 2013; Nandan *et al.*, 2019a,b). The approach is compatible with a range of models used operationally, including epidemic-type aftershock sequence (ETAS) model and its variants (Werner *et al.*, 2011; Zhuang, 2011; Hainzl *et al.*, 2024; Mizrahi *et al.*, 2024; Han *et al.*, 2025).

Three methodological issues arise when estimating the full PDF $P(n)$:

1. Randomness: Because $P(n)$ is typically derived from a large but finite ensemble of simulated catalogs, sampling variability introduces artificial roughness into the distribution. To mitigate this effect, studies have traditionally adopted parametric count models, such as Poisson; negative-binomial or beta-negative binomial approximations; or, more recently, nonparametric smoothers such as adaptive kernel density estimators (Kagan, 2010, 2017; Werner *et al.*, 2011; Harte, 2017; Nandan *et al.*, 2019a; Taroni *et al.*, 2023; Molchan and Varini, 2024). The use of the Poisson distribution is particularly natural when a forecast specifies only the mean rate because under the maximum entropy principle, the Poisson distribution is the most parsimonious choice: it imposes the fewest assumptions beyond the mean number of events. However, when a forecast conveys information beyond the mean rate, restricting the distribution to Poisson is no longer theoretically justified. In this context, nonparametric approaches, especially adaptive kernel density estimators, have emerged as effective tools for stabilizing likelihood-based evaluations while minimizing model-induced bias

(Nandan *et al.*, 2019a; Nandan, Kamer, *et al.*, 2021; Nandan, Ram, *et al.*, 2021).

2. Fat-tailed (or power law-tailed) behavior: The frequency distribution of counts across cells is empirically close to a power law $P(n) \sim n^{-(1+a)}$ in its upper tail, with an exponent a often smaller than 2, qualifying the terminology “heavy tailed” (Embrechts *et al.*, 1997; Sornette, 2006). The term “fat tailed” for distributions is broader and refers to distributions that have right-end tails that are thicker than those of Gaussian and exponential distributions. To distinguish with the term “heavy tail,” the term “fat-tailed distribution” may usually refer to a power law with exponent a larger than 2, in which case the variance is finite, or it may have a nonpower law tail such as a stretched exponential (Laherrère and Sornette, 1998). We shall use the term “fat-tailed” in the broader sense.

The power law tail of the PDF $P(n)$ reflects clustered triggering and cascade dynamics (Kagan, 2017; Nandan *et al.*, 2019a; Nandan, Kamer, *et al.*, 2021; Fig. 1b in the present study for an illustration). This power law property is in fact predicted by theory for ETAS models of earthquake triggering (Saichev *et al.*, 2005; Saichev and Sornette, 2006, 2007). Yet most evaluation pipelines, especially those relying on nonparametric smoothing, do not explicitly constrain or extrapolate the tail (Nandan *et al.*, 2019a; Nandan, Ram, *et al.*, 2021; Mizrahi *et al.*, 2024; Han *et al.*, 2025).

3. Structural limitations of ETAS-based simulations: In addition to the statistical fat-tailed behavior of earthquake-count distributions, out-of-range observations may also reflect a more fundamental limitation of standard ETAS formulations. In cells impacted by aftershocks of great earthquakes, observed counts can exceed the maximum of the simulated ensemble, requiring an out-of-range tail treatment (e.g., extrapolation versus truncation). How such tail handling affects comparative model ranking has not been systematically quantified. Although ETAS models are effective in describing short-term triggering and aftershock cascades, they generally assume a background seismicity rate that is stationary in time and space. They do not explicitly incorporate long-term tectonic loading, fault-specific recurrence behavior, or time-dependent stress accumulation. As a result, large but physically plausible earthquakes that are absent from the calibration catalog may be assigned extremely low probability or may be effectively missing from the simulated ensemble. This limitation is particularly relevant for fault systems such as the Longmenshan fault zone, where the 2008 M_s 8.0 Wenchuan earthquake and its aftershock sequence strongly reshaped the seismicity pattern in the adjacent region. Therefore, the out-of-range problem should not be interpreted solely as a numerical artifact of finite simulations or as a consequence of fat-tailed count distributions. It also highlights the structural incompleteness of purely ETAS-based forecasting models when

rare, fault-controlled large earthquakes are not represented in the calibration history.

Increasing the number of simulated catalogs can, to some extent, mitigate these three issues mentioned. However, simply enlarging the ensemble size is not a practical solution in real applications. Doing so would not only lead to a substantial waste of computational resources but also significantly prolong the running time, thereby raising the overall computational cost. More importantly, even an arbitrarily large number of simulated catalogs cannot fully guarantee a perfect resolution of the underlying problems because the fundamental limitations are methodological rather than merely statistical.

This study addresses that gap. We develop an evaluation framework that (1) estimates the cell-wise probability distribution $P(n)$ of earthquake numbers n with adaptive smoothing, (2) explicitly incorporates fat-tail behavior with principled extrapolation for out-of-range observations, and (3) tests how sensitive model ranking is to these choices. Our analysis complements recent advances in short-term forecasting across diverse regions and model classes, including ETAS generalizations and alternative point-process formulations (Werner *et al.*, 2011; Zhuang, 2011; Davis *et al.*, 2024; Hainzl *et al.*, 2024; Mizrahi *et al.*, 2024; Han *et al.*, 2025).

To enable transparent and repeatable testing at scale, we introduce AoyuX, an online research platform dedicated to earthquake predictability studies in China. Developed by the Institute of Risk Analysis, Prediction, and Management (Risks-X) at the Southern University of Science and Technology (SUSTech) with the China Earthquake Networks Center (CENC) and other institutions as partner organizations, AoyuX integrates state-of-the-art statistical seismology, centered on ETAS-type models that jointly forecast time–space–magnitude probabilities, with an extensible architecture for benchmarking, visualization, and (future) model-comparison challenges. The initial operational focus is the China Seismic Experimental Site (CSES) in Sichuan–Yunnan region, one of the nation’s most instrumented and seismically active regions (Wu *et al.*, 2019; Wu and Li, 2021a,b; Li *et al.*, 2022; Wu, 2022). Since June 2024, AoyuX has issued monthly, quarterly, semiannual, and annual forward forecast bulletins for the CSES (see [Data and Resources](#)), thereby establishing a sustained prospective and pseudoprospective testing environment. Looking ahead, AoyuX will fuse geophysical observations with artificial intelligence–assisted and physics-based components (e.g., Zhan *et al.*, 2024; Zhang *et al.*, 2025) to support mid-term, short-term, and imminent forecasting and to facilitate communication between experts and the public.

In the present study, we conducted an ~25-yr month-by-month pseudoprospective forecasting experiment for the CSES using the ETAS model together with a spatially heterogeneous variant in which the background seismicity rate varies across space. We further evaluated the impact of innovative

fat-tail treatments for out-of-range observations and investigated under which conditions the cumulative information gain (CIG) curves exhibit a clear long-term upward trend, or alternatively, when such a trend is obscured by stochastic fluctuations. We also examined the spatial distribution of mean information gain (MIG) to characterize where the ETAS model with spatially heterogeneous background seismicity differs from its homogeneous background counterpart. By integrating a fat tail–aware scoring methodology with a dedicated open testing platform, our work advances both the methodological foundations and the research infrastructure required for fair and statistically rigorous assessment of earthquake forecasting models in China and beyond.

Methods

ETAS model for earthquake forecasting

The ETAS model is a widely used formulation of the self-exciting Hawkes point process, specifically adapted to characterize seismicity. Unlike Markovian processes, the ETAS model generally exhibits non-Markovian behavior when its temporal memory kernel deviates from an exponential form (Hawkes and Oakes, 1974; Kagan and Knopoff, 1981, 1987; Ogata, 1988, 1998; Kanazawa and Sornette, 2020; Kanazawa and Sornette, 2024). At any spatiotemporal location (x, y, t) , the model defines the expected seismicity rate or intensity function $\lambda(x, y, t)$, which depends on the complete history of seismic events up to time t , denoted as H_t :

$$\begin{aligned}\lambda(x, y, t | H_t) &= \mu + \sum_{i: t_i < t} F(m_i) T(t - t_i) S(x - x_i, y - y_i, m_i) \\ &= \mu + \sum_{i: t_i < t} K e^{\alpha(m_i - M_{co})} \frac{T_{\text{norm}} e^{-\frac{t-t_i}{\tau}}}{(t - t_i + c)^{1+\omega}} \\ &\quad \times \frac{S_{\text{norm}}}{[(x - x_i)^2 + (y - y_i)^2 + d e^{\gamma(m_i - M_{co})}]^{1+\rho}}.\end{aligned}\quad (1)$$

Here, the intensity λ consists of a background rate μ and contributions from all prior earthquakes. Each past event i , characterized by its magnitude m_i and location (x_i, y_i, t_i) , influences future seismicity through a separable spatiotemporal triggering kernel, weighted by a fertility (or productivity) function $F(m_i)$, parameterized by constants K and α . The spatiotemporal triggering kernel is typically defined as the product of two components: a temporal kernel described by the Omori–Utsu law $T(t - t_i)$, with parameters c , ω , and τ and a spatial kernel represented by a Green’s function with parameters d , γ , and ρ together describing the temporal and spatial dependencies of aftershock triggering. The normalization constants T_{norm} and S_{norm} ensure that the temporal and spatial kernels are valid PDFs. This formulation reflects a commonly adopted choice for the spatiotemporal triggering kernel in ETAS models (Nandan, Ram, *et al.*, 2021; Nandan *et al.*, 2022), although alternative kernel forms have also been proposed in the

literature (Felzer *et al.*, 2002, 2003; Helmstetter and Sornette, 2002; Zhuang *et al.*, 2002, 2005).

The ETAS model admits two conceptually distinct yet mathematically equivalent interpretations (Helmstetter and Sornette, 2002; Li, Sornette, Wu, *et al.*, 2025):

1. Branching interpretation (Hawkes and Oakes, 1974; Kagan, 1991; Sornette and Werner, 2005b): Each observed earthquake can be probabilistically linked to a parent event, forming a cascade of primary and secondary aftershocks in a tree-like structure. In this framework, background events follow a Poisson process, and each event may trigger its own sequence of aftershocks. Every tree configuration is associated with a probability that quantifies the likelihood of a specific parent–offspring association, given the observed catalog. These probabilities are derived from the ETAS intensity function $\lambda(x, y, t)$ (Zhuang *et al.*, 2002).
2. Collective influence interpretation (Helmstetter and Sornette, 2002): Each nonbackground event arises from the cumulative triggering effect of all preceding earthquakes. In this view, the intensity function $\lambda(x, y, t)$ represents the combined influence of historical seismicity at a given location and time, superimposed on the background rate.

The mathematical equivalence between these two interpretations was rigorously established by Hawkes and Oakes (1974). Sornette and Werner (2005b) further offered an intuitive explanation for this equivalence, based on the combination of (1) the formulation of the ETAS seismicity rate expressed as a linear superposition of contributions from prior events and of (2) the exponential form of the point process hazard rate. This dual perspective enables the ETAS model to function both as a generative engine of synthetic earthquake catalogs and as a predictive tool for time-dependent seismic hazard assessment.

In the present study, we employ two variants of the ETAS model that share an identical triggering kernel, with the sole difference being the treatment of the background seismicity rate (Nandan, Ram, *et al.*, 2021; Nandan *et al.*, 2022):

- I) An ETAS model with a constant background rate (ETAS _{μ}). This variant, defined by equation (1), assumes a spatially homogeneous background rate μ . We adopt it as the benchmark model in the present study. Its estimation involves determining the following set of model parameters: $\{\mu, K, \alpha, c, \omega, \tau, d, \gamma, \rho\}$.
- II) An ETAS model with spatially variable background rate (ETAS _{$\mu(x, y)$}). In this augmented variant, the background rate $\mu(x, y)$ is treated as a nonparametric function that captures spatial heterogeneity of the background events. Its estimation follows a weighted kernel approach, as introduced by Nandan, Ram, *et al.* (2021) and Nandan *et al.* (2022).

$$\mu(x, y) = \frac{1}{T} \sum_{i=1}^{N_{\text{eq}}} IP_i \cdot \frac{QD^{2Q}}{\pi} \frac{1}{[(x - x_i)^2 + (y - y_i)^2 + D^2]^{1+Q}}, \quad (2)$$

in which the summation is over all N_{eq} earthquakes in the catalog, weighted by their probability IP_i of being background events (Zhuang *et al.*, 2002). The normalization by the catalog duration T ensures that $\mu(x, y)$ represents the seismicity rate per unit time, and the factor QD^{2Q}/π normalizes the kernel per unit area at location (x, y) . The choice of a power law kernel is motivated by its demonstrated superiority over the commonly used Gaussian kernel in previous studies (e.g., Helmstetter *et al.*, 2007; Nandan, Kamen, *et al.*, 2021). Its estimation involves determining the following set of parameters: $\{Q, D, K, \alpha, c, \omega, \tau, d, \gamma, \rho\}$.

The parameter calibration of both variants is carried out using the extended expectation–maximization (EM) algorithm (Veen and Schoenberg, 2008). We refer readers to Nandan, Ram, *et al.* (2021) and Nandan *et al.* (2022) for full methodological details.

Pseudopropective forecasting experiments

The pseudopropective forecasting experiment is conducted over the testing period from 1 January 2000 to 31 December 2023. Each forecast window spans one calendar month, and the windows are nonoverlapping. Specifically, for a given forecast window (e.g., a particular month in a given year), the earthquake catalog prior to that month is used to calibrate the model parameters. Based on the calibrated parameters, 100,000 synthetic earthquake catalogs are simulated for the forecast window. These synthetic catalogs are used to estimate the occurrence probability of the future earthquakes. The process is then repeated for each subsequent month in the testing period. To assess the spatial performance of the models, the study region is discretized using a geodesic triangular tessellation on the Earth’s surface accounting for its spherical geometry, with six different spatial resolutions defined by equal-area triangles of $\sim 195, 779, 3,117, 12,470$, and $49,883 \text{ km}^2$ and the entire CSES region as a single unit ($\sim 780,000 \text{ km}^2$). In addition, five different magnitude thresholds M_t are considered: 3.0, 3.5, 4.0, 4.5, and 5.0.

For each forecast window, magnitude threshold, and spatial resolution (space–time–magnitude bin), we compute the distribution of the number of simulated earthquakes in each triangular cell. The probability distribution $P(n)$, in which n denotes the number of earthquakes, is then estimated using the kernel-based smoothing method described in the next section. Accordingly, the probability of observing n_i earthquakes in the i th triangle is given by $P(n_i)$. Given a particular spatial resolution and magnitude threshold, the LL score of a model for a specific forecast window is defined as

$$LL = \sum_{i=1}^N \log P(n_i), \quad (3)$$

in which N is the number of spatial cells at the specified resolution. The information gain (IG) is defined as the difference in LL scores between $ETAS_{\mu(x,y)}$ and $ETAS_{\mu}$. A higher IG indicates superior performance of the spatially heterogeneous $ETAS_{\mu(x,y)}$ model relative to the benchmark $ETAS_{\mu}$ model and vice versa.

Distribution of forecasted earthquake numbers

To obtain a smooth distribution of forecasted earthquake numbers and minimize uncertainties arising from the finite number of simulations, we adopt an adaptive Gaussian kernel density estimation (KDE) method to represent the distribution of simulated earthquake numbers within each grid cell, given a specific forecast window, magnitude threshold M_t , and spatial resolution. The adaptive KDE method is a nonparametric approach that adjusts to varying data density by allowing the bandwidth to vary locally (Cranmer, 2001). Its key characteristic is that regions with higher data density are assigned narrower kernels, whereas sparser regions use wider kernels (Fig. 1a), thereby improving estimation accuracy across the full distribution.

Although earthquake counts are discrete nonnegative integers, we use the adaptive Gaussian KDE here as a smoothing device for the empirical distribution obtained from a large but finite ensemble of simulated catalogs rather than as an assumption that the underlying count process is continuous. In practice, the KDE-smoothed density is evaluated only at the observed integer count n_i and is used as a regularized KDE-based likelihood score rather than being interpreted literally as the discrete probability mass $P(N = n_i)$. Because $\hat{f}(n_i)$ is a continuous density value rather than a probability mass, its use here can be understood as a unit-width approximation to the corresponding discretized probability mass. Formally, this mass would be $\left[\int_{n_i-1/2}^{n_i+1/2} \hat{f}(x) dx \approx \hat{f}(n_i) \right]$ with appropriate boundary treatment at $n_i = 0$. This treatment is particularly useful when the number of synthetic catalogs is finite because the raw empirical histogram can contain artificial gaps or strong sample-to-sample fluctuations (finite-simulation noise), especially in the upper tail, where the number of simulated realizations is small. The continuous Gaussian kernel therefore provides a practical way to stabilize the KDE-based likelihood calculation while retaining the nonparametric character of the simulated forecast distribution.

Discrete alternatives, such as discrete KDE or parametric count models including Poisson, negative-binomial, and beta-negative-binomial distributions, are possible in principle. However, these approaches would introduce additional assumptions about the functional form of $P(n)$. In particular, a Poisson model is appropriate mainly when the forecast specifies only the mean rate, whereas the simulated ETAS catalogs contain information about the full distribution of earthquake counts,

including clustering-induced overdispersion and fat-tailed behavior. Negative binomial-type models can represent overdispersion, but they may still be too restrictive for the irregular and model-specific distributions produced by aftershock cascades. We therefore adopt adaptive Gaussian KDE as a minimally parametric compromise that smooths the simulated distribution without forcing it into a prescribed count-family form. In our tests, using alternative smoothing methods does not materially change the results.

Smoothing can in principle affect the KDE-based likelihood score and hence model ranking. If the distribution is over-smoothed, sharp but meaningful differences between models may be attenuated; if it is undersmoothed, finite-simulation noise may be amplified and spuriously influence the LL. In the present study, this effect is mitigated in three ways. First, the same adaptive KDE procedure is applied consistently to both $ETAS_{\mu}$ and $ETAS_{\mu(x,y)}$ under each time-space-magnitude configuration. Second, the adaptive bandwidth reduces noise in sparse regions while preserving narrower kernels in high-density regions. Third, cases in which the observed count lies outside the simulated range are handled separately through the three tail-treatment strategies described subsequently. Thus, the KDE mainly regularizes within-range KDE-based likelihood scores, whereas the sensitivity of model ranking to extreme tail treatment is explicitly evaluated in the [CIG of \$ETAS_{\mu\(x,y\)}\$ over \$ETAS_{\mu}\$](#) and [Bias from improper tail treatment](#) sections.

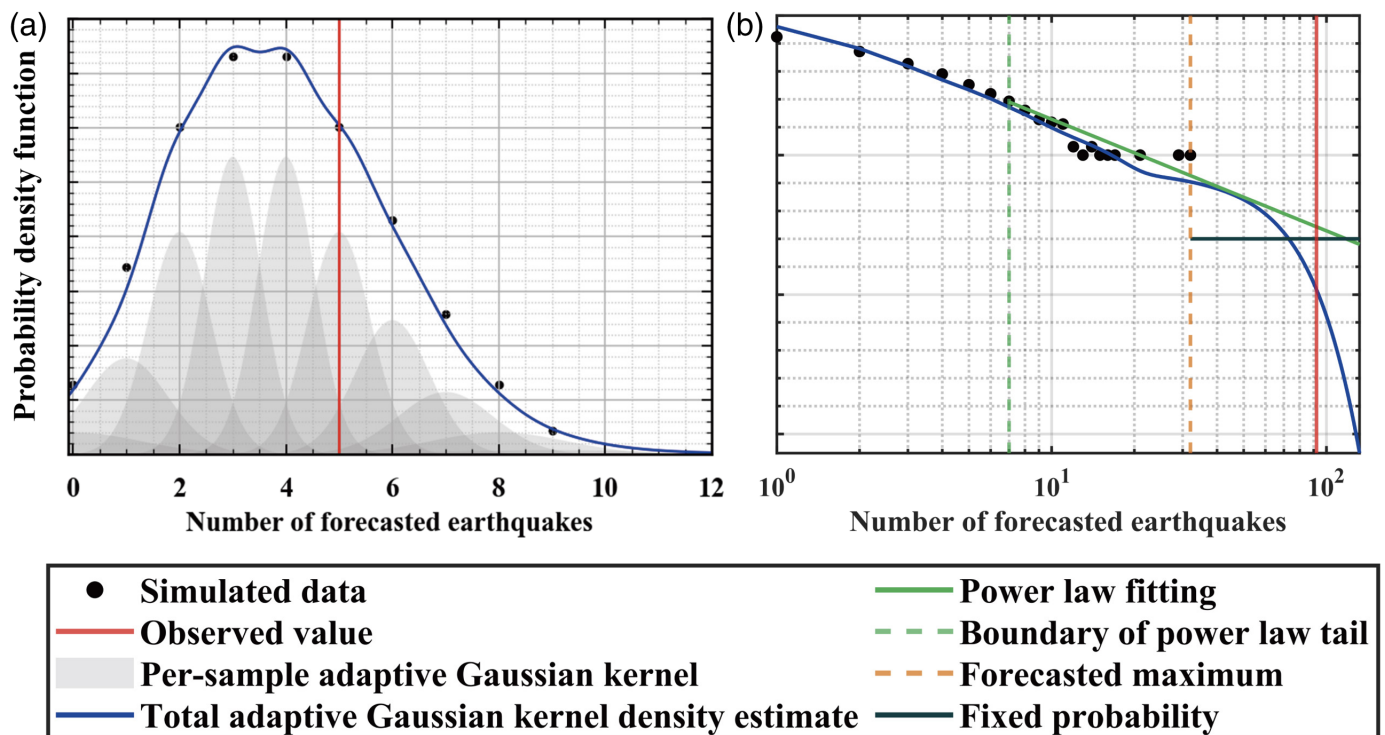
In the present study, the kernel density $f(n)$ for the number of simulated earthquakes n is given by

$$f(n) = \frac{1}{N_{\text{sim}}} \sum_{j=1}^{N_{\text{sim}}} \frac{1}{h_j} K\left(\frac{n - n_j}{h_j}\right), \quad (4)$$

in which N_{sim} is the number of synthetic catalogs (set to 100,000 in the present study), n_j is the number of earthquakes in the j th simulated catalog, $K(\cdot)$ is the standard Gaussian kernel with zero mean and unit variance, and h_j is the locally adaptive bandwidth for the j th sample. To compute the pilot kernel density $f_0(n)$, each h_j is initially set to a fixed bandwidth h_0 , which is selected to minimize the mean integrated square error following a rule-of-thumb bandwidth selection (Scott, 2015). The adaptive bandwidth h_j associated with the simulated count n_j is then determined using the classical square-root law:

$$h_j = \left[\frac{h_0}{f_0(n_j)g} \right]^{-1/2}, \quad (5)$$

where $f_0(n_j)$ is the pilot density estimate evaluated at n_j , and g is the geometric mean of the pilot density estimates. This formulation yields narrower bandwidths in regions where the simulated earthquake counts are concentrated and broader bandwidths in sparsely populated regions, while preserving h_0 as the global reference bandwidth. In effect, each simulation



contributes a Gaussian kernel in probability space, with its width adaptively scaled by the local data density, that is, the number of synthetic catalogs yielding similar outcomes.

However, when the observed number of earthquakes within a grid cell—given a specific forecast window, magnitude threshold M_t , and spatial resolution—exceeds the maximum value obtained from the simulated catalogs, the adaptive kernel method may yield serious biased estimates of the probability $P(n)$ (see, e.g., Fig. 1 in the present study and fig. 1 in Nandan *et al.*, 2019a). To address this issue, we propose three alternative strategies for assigning $P(n)$ in such cases (Fig. 1b):

1. Power law extrapolation: Fit a power law function to the upper 30% of the simulated tail data and use the extrapolated value as $P(n)$.
2. Fixed minimum probability: Assign a fixed probability value equal to $1/(N_{\text{sim}})^2$.
3. Exclusion: Exclude the corresponding $P(n)$ value entirely from the likelihood computation.

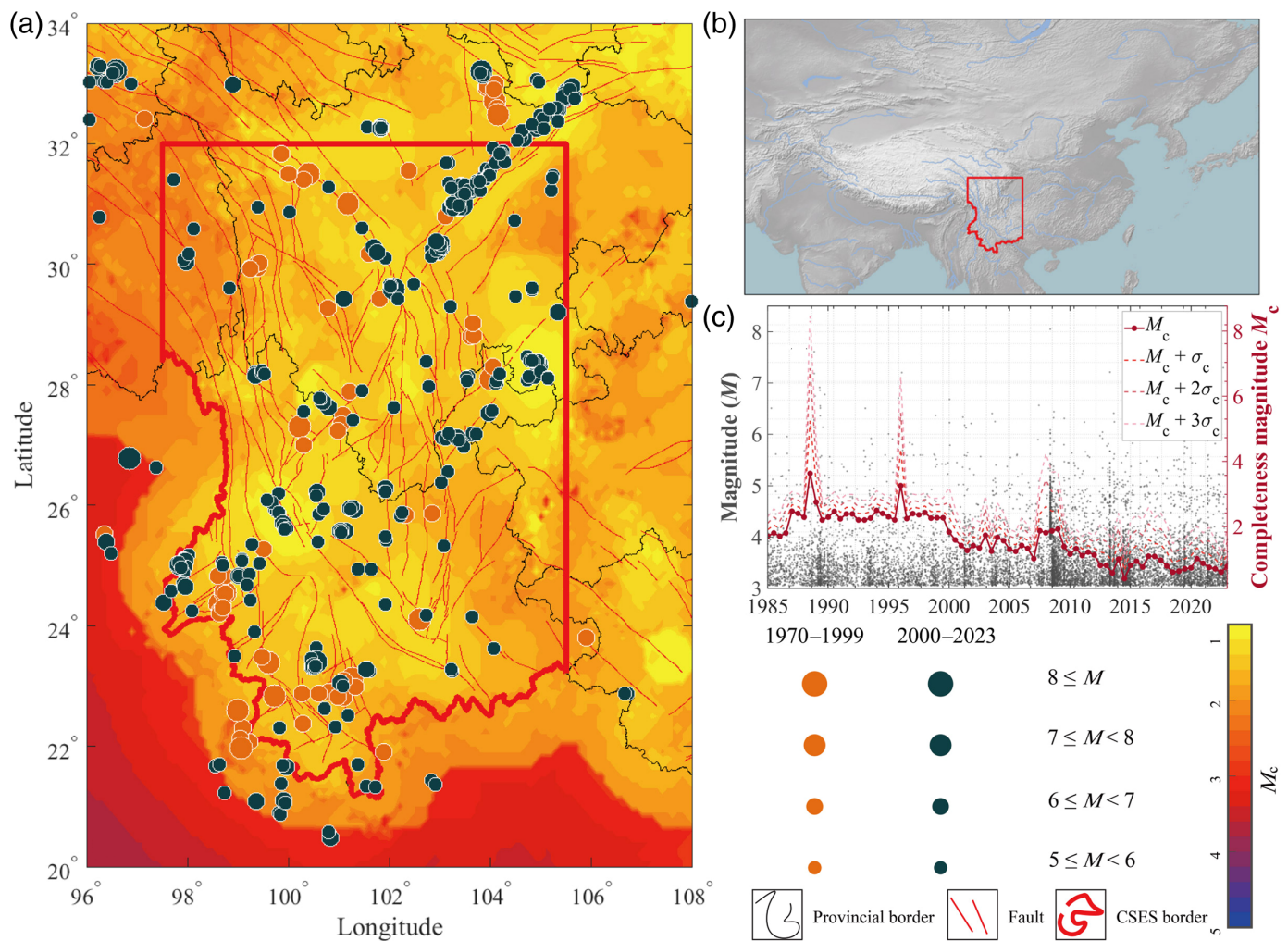
In the following sections, we evaluate the impact of these three strategies on model performance in forecasting experiments.

Seismicity in CSES

Situated in a seismically active region (Li *et al.*, 2020; 2021; 2022), the CSES is designed to integrate dense, multidisciplinary observations to investigate the complex physical processes underlying earthquake generation (Wu *et al.*, 2019; Wu and Li, 2021a,b; Wu, 2022). By combining geological, geophysical, and geochemical data, the site enables a comprehensive investigation of seismic hazards. This integrative approach facilitates

Figure 1. Illustration of the probability distribution for the number of forecasted earthquakes, shown for two scenarios: (a) when the observed value (red solid line) falls within the range of simulated values and (b) when the observed value exceeds the maximum of the simulated distribution. The forecasted probability corresponding to the observed value n_i in the i th triangle is denoted as $P(n_i)$. In the present study, simulated data (solid black circles) are generated from N_{sim} sets of synthetic catalogs ($N_{\text{sim}} = 100,000$). The per-sample adaptive Gaussian kernels (gray-shaded areas), each with its own bandwidth, are summed to construct the total adaptive Gaussian kernel density estimate (solid blue line), which is used to present the overall probability distribution. When n_i (solid red line) exceeds the forecasted maximum (dashed orange line) in panel (b), we propose three alternative strategies for assigning $P(n_i)$: (1) fitting a power law function (solid green line) to the upper 30% of the simulated tail data (to the right of the dashed green line); (2) assigning a fixed probability value (horizontal black line), equal to $1/(N_{\text{sim}})^2$; or (3) excluding $P(n_i)$ altogether in such cases. The color version of this figure is available only in the electronic edition.

the development and validation of predictive models, thereby contributing significantly to earthquake forecasting efforts in China. The CSES region has long maintained a dense and high-quality seismic monitoring network, resulting in an extended record of reliable earthquake catalogs (Li *et al.*, 2023; Li, Sornette, and Feng, 2025). As shown in Figure 2, the spatiotemporal distribution of completeness magnitude M_c indicates that the CSES catalog has been nearly complete for events with $M \geq 3$ since 1985. Short-lived degradations of completeness are observed mainly around several strong earthquake sequences, especially around 1988 and 1995 and following the 2008 M_s 8.0 Wenchuan earthquake, when the

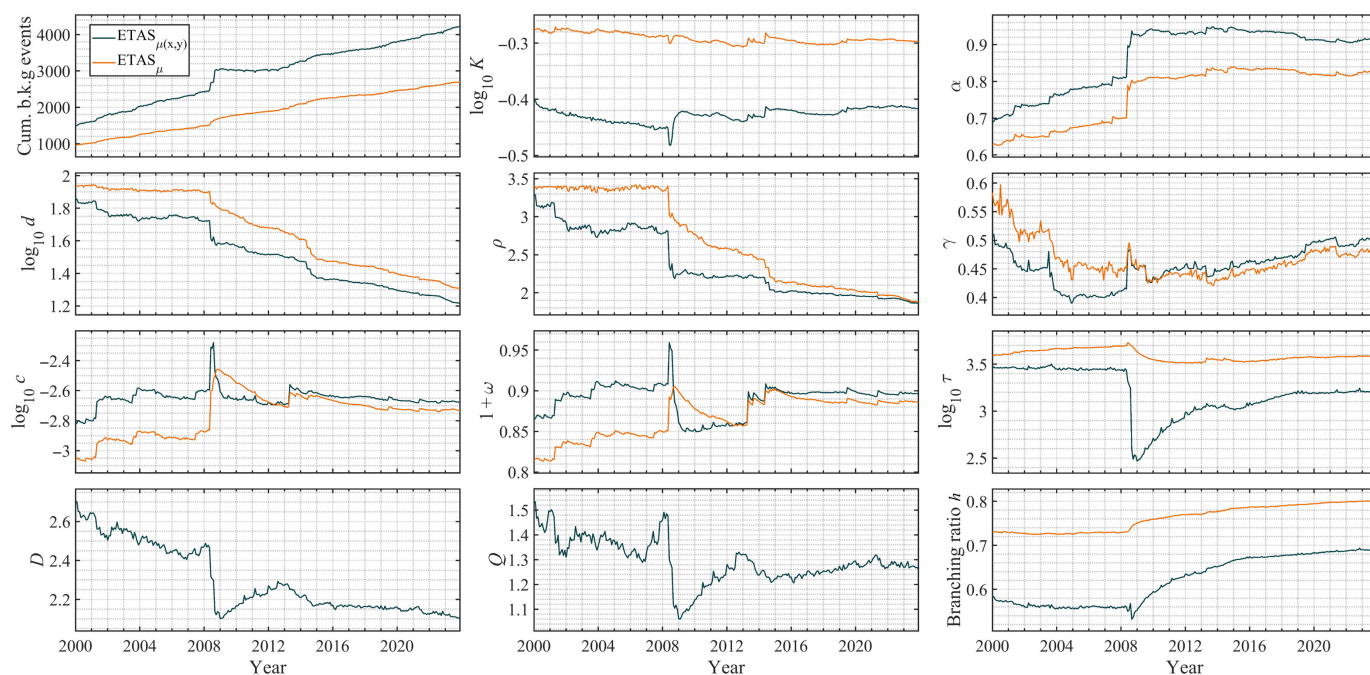


completeness probability for $M \geq 3$ dropped to $\sim 50\%$ to 97.8% (Fig. 2c). These transient episodes are typical of short-term aftershock incompleteness caused by extremely high aftershock rates and overlapping seismic signals immediately after large earthquakes. Therefore, although the catalog is not perfectly complete at all times, the use of 1985 as the starting time for ETAS calibration does not introduce a persistent completeness bias at the adopted catalog cut-off magnitude at $M_{co} = 3.0$. Instead, it provides a longer historical record for constraining the background seismicity rate and the triggering history prior to the testing period.

Nevertheless, short-term incompleteness after strong earthquakes can affect ETAS parameter estimation and should be acknowledged. Missing early aftershocks may reduce the apparent number of triggered events, thereby biasing the estimated triggering productivity, branching ratio, and early-time Omori-Utsu decay parameters. In particular, the productivity-related parameters K and α , as well as the temporal kernel parameters c and ω , can be affected because the earliest and densest part of an aftershock sequence is undersampled. At the same time, missing events can break parent-offspring links in the inferred triggering structure, causing some events that are physically triggered to be classified as independent

Figure 2. Spatiotemporal distribution of seismicity and completeness magnitude (M_c) in the China Seismic Experimental Site (CSES), located in the Sichuan–Yunnan region. (a) Spatial distribution of earthquakes from 1970 to 2023, completeness magnitude (M_c) since 2009, and active faults within the CSES. The background map of M_c is adapted from Li *et al.* (2023). Fault locations are taken from Deng *et al.* (2003). (b) Location of the CSES shown on a broader regional map. (c) Temporal evolution of M_c (right y axis) and the magnitude–time (MT) plot of earthquakes with $M \geq 3$ (left y axis) since 1985. The estimation of M_c was performed using a half-year step and one-year window. The dashed red lines with decreasing color indicate completeness probabilities of 84.2% ($M_c + \sigma_c$), 97.8% ($M_c + 2\sigma_c$), and 99.9% ($M_c + 3\sigma_c$), respectively. M_c and σ_c are estimated by the method proposed by Wang *et al.* (2025) and Li *et al.* (2026). The earthquake magnitudes shown in the magnitude–time plot have been perturbed by a uniformly distributed random error within the range $[-0.05, 0.05]$. The color version of this figure is available only in the electronic edition.

background events. This may lead to an apparent increase in the estimated background component, especially for the spatially variable background rate $\mu(x,y)$. Such effects have been discussed extensively in ETAS-based studies of incomplete catalogs and short-term aftershock incompleteness (Sornette and Werner, 2005b; Omi *et al.*, 2014; Zhuang



et al., 2017; Mizrahi *et al.*, 2021; Hainzl, 2022; Hainzl *et al.*, 2024). In the present study, these effects are expected to be temporally localized around a few major earthquake sequences and may partly explain the abrupt parameter changes observed in Figure 3, but they do not undermine the overall use of $M_{co} = 3.0$ for the long-term calibration catalog.

The earthquake catalogs used in the present study were provided by the CENC. The catalog of earthquakes with $M \geq 3.0$ in the CSES region during 1970–2023 is available as Data Set S1 in Li, Sornette, Wu, *et al.* (2025). The CSES catalog includes a mixture of magnitude types, primarily local magnitude (M_L) and surface-wave magnitude (M_s), which introduces potential complications because of magnitude-type heterogeneity. To evaluate the possible influence of this issue, we performed a simple sensitivity test by varying the relative contributions of different magnitude types in a pseudoprospective forecasting experiment. The results indicate that magnitude-type heterogeneity may introduce some variability, but it does not systematically alter the overall CIG patterns or dominate the model comparison results. The magnitude bin size is set to 0.1. Temporally, the catalog is divided into two segments: (1) an auxiliary catalog from 1 January 1970 to 31 December 1984, used to avoid misclassifying early events in the primary catalog as background events in the ETAS model, and (2) the primary catalog, extending from 1 January 1985 to the day before the forecast window, used for calibrating the ETAS model parameters.

Results

In this section, we present two sets of results: (1) the calibrated parameters of the ETAS models and (2) the outcomes of pseudoprospective earthquake forecasting experiments based on the calibrated models.

Figure 3. Temporal evolution of the calibrated parameters in the epidemic-type aftershock sequence (ETAS) model with spatially variable background seismicity ($ETAS_{\mu(x,y)}$) and in the ETAS model with a constant background rate ($ETAS_{\mu}$). For model calibration, the primary catalog spans from 1 January 1985 to the day before each monthly forecast window and is preceded by a 15-yr auxiliary catalog (from 1 January 1970 to 31 December 1984) to prevent misclassification of early events in the primary catalog as background events. Note that the parameters D and Q are specific to $ETAS_{\mu(x,y)}$ and are not included in $ETAS_{\mu}$. The color version of this figure is available only in the electronic edition.

Calibrated parameters of the ETAS model

To examine the temporal stability and evolution of ETAS model parameters, we calibrate both $ETAS_{\mu(x,y)}$ and the homogeneous counterpart $ETAS_{\mu}$ over a series of monthly forecast windows. For each calibration step, the primary catalog includes all events from 1 January 1985 up to the day preceding the forecast window. In addition, a 15-yr auxiliary catalog covering the period from 1 January 1970 to 31 December 1984 is appended to provide historical context and reduce the likelihood of misclassifying earlier events as background earthquakes.

Figure 3 shows the temporal evolution of the estimated parameters for both $ETAS_{\mu(x,y)}$ and $ETAS_{\mu}$. Overall, the two models yield noticeably different parameter estimates, with $ETAS_{\mu(x,y)}$ identifying significantly more background events than $ETAS_{\mu}$, primarily as a result of allowing for a spatially heterogeneous background seismicity, which is otherwise likely to be misclassified as triggered seismicity in $ETAS_{\mu}$. Furthermore, from 2000 to 2023, the differences between the two models in several key parameters remain relatively stable over time. Specifically, the fertility law parameters ($\log_{10} K$

and α), the spatial kernel parameter ($\log_{10} d$), and the branching ratio h consistently exhibit similar difference amplitudes between $\text{ETAS}_{\mu(x,y)}$ and ETAS_{μ} . In contrast, the gap between the two models in terms of the cumulative number of background events and $\log_{10} \tau$ gradually widens over time. On the other hand, the differences in ρ , γ , $\log_{10} c$, and $1 + \omega$ tend to decrease with time and, in some cases, nearly vanish. Notably, abrupt changes in several parameter time series coincide with the occurrence of major earthquakes in the region. Examples include the 2008 M_s 8.0 Wenchuan earthquake (12 May), the 2013 M_s 7.0 Lushan earthquake (20 April), and the 2017 M_s 7.0 Jiuzhaigou earthquake (8 August).

CIG of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ}

The calibrated parameters prepared for each forecast window were used to generate synthetic earthquake catalogs based on the first of the two conceptually distinct yet mathematically equivalent interpretations of the ETAS model, that is, the branching interpretation (see the second paragraph of The [ETAS model for earthquake forecasting](#) section). Specifically, for each forecast window, the ETAS parameters calibrated using the catalog prior to that window were used to simulate the seismicity expected during the target month. The simulated seismicity in each forecast window consists of three components: earthquakes triggered by events that occurred before the beginning of the forecast window, background earthquakes generated during the forecast window, and earthquakes triggered by these background events within the same window. In this way, the triggering memory inherited from the prewindow catalog is explicitly propagated into the forecast period, while newly generated background events are also allowed to initiate their own aftershock sequences. To reduce boundary effects in both model calibration and simulation, we used an auxiliary region extending 100 km beyond the CSES boundary; however, only earthquakes occurring within the CSES region were retained for forecast evaluation. The b value used for magnitude sampling was treated in the same way as the other ETAS parameters: for each forecast window, it was estimated from the catalog prior to that window and then used in the corresponding simulations. For each forecast window, 100,000 synthetic catalogs were generated in this study. All simulated catalogs produced by the two ETAS models are openly available through [Li, Zhang, and Sornette \(2025\)](#).

Figure 4 presents the time series of CIG of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ} from 2000 to 2023. Results are shown for six spatial resolutions, five magnitude thresholds (M_t), and three strategies for assigning $P(n_i)$ when the observed value n_i in the i th triangle exceeds the maximum value obtained from the simulations. The green, black, and orange curves correspond to strategies (1), (2), and (3), respectively, described in the [Distribution of forecasted earthquake numbers](#) section. We also provide, in the [Data and Resources](#) section, adaptive Gaussian kernel density fits for both models across the five magnitude thresholds for the full CSES region.

Table 1 reports the percentage of out-of-range grid cells across all 288 forecast windows for each combination of spatial resolution and magnitude threshold M_t . For the gridded spatial resolutions, the percentages are generally low at $M_t = 3$ and $M_t = 3.5$, but these configurations contain a large number of spatial cells, so the absolute number of out-of-range cases is still nonnegligible. The percentage increases markedly at higher magnitude thresholds, especially for $M_t = 4.5$ and $M_t = 5$, where the forecasted count distributions become increasingly sparse. These results show that out-of-range observations are not restricted to a single extreme event but occur across different space–magnitude configurations, particularly in sparse-count settings. This motivates the comparison of the three tail-treatment strategies in Figure 4.

For $M_t = 3$ and 3.5 (i.e., the first and second columns of Fig. 4 across all spatial resolutions) and for the entire CSES region (i.e., the last row of Fig. 4 across all M_t), the CIG curves of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ} remain consistently above zero and exhibit an overall upward trend over time. This indicates that $\text{ETAS}_{\mu(x,y)}$ outperforms the baseline ETAS_{μ} model in terms of forecasting skill under these settings. For other combinations of spatial resolution and M_t , however, the results are more ambiguous. The CIG curves often display substantial fluctuations (see the discussion in the [Quantitative rationale for the variability of CIG](#) section), and in many cases, the performance of $\text{ETAS}_{\mu(x,y)}$ does not significantly exceed that of ETAS_{μ} , reflecting the complex nature of seismicity in the study region. As with the parameter time series discussed in the [Calibrated parameters of the ETAS model](#) section, many abrupt changes in CIG coincide with major earthquakes in the region, most notably the 2008 M_s 8.0 Wenchuan earthquake (12 May), which is discussed in detail in the [An example of out-of-range earthquake counts: The May 2008 Wenchuan earthquake](#) section.

Figure 4 also shows that the choice of strategy for assigning $P(n_i)$ when the observed value n_i exceeds the forecasted maximum can significantly influence the model evaluation results. For lower magnitude thresholds (e.g., $M_t = 3$ –4), corresponding to the left side panels in Figure 4, the three strategies—power law extrapolation, fixed minimum probability, and exclusion—yield markedly different CIG curves. This highlights the sensitivity of model evaluation to how such rare cases are handled. In most instances, the power law extrapolation strategy results in the highest CIG values. However, there are exceptions; for example, under the combinations of spatial resolution 195, 3117, 12,470, and 49,883 with $M_t = 3$, the fixed minimum probability strategy consistently produces higher CIG curves than the power law extrapolation approach.

Bias from improper tail treatment

However, the apparent advantage of the CIG curves obtained using the fixed minimum probability strategy for assigning $P(n_i)$ is in fact illusory. When both models ($\text{ETAS}_{\mu(x,y)}$ and ETAS_{μ}) assign $P(n_i)$ based on the same fixed minimum

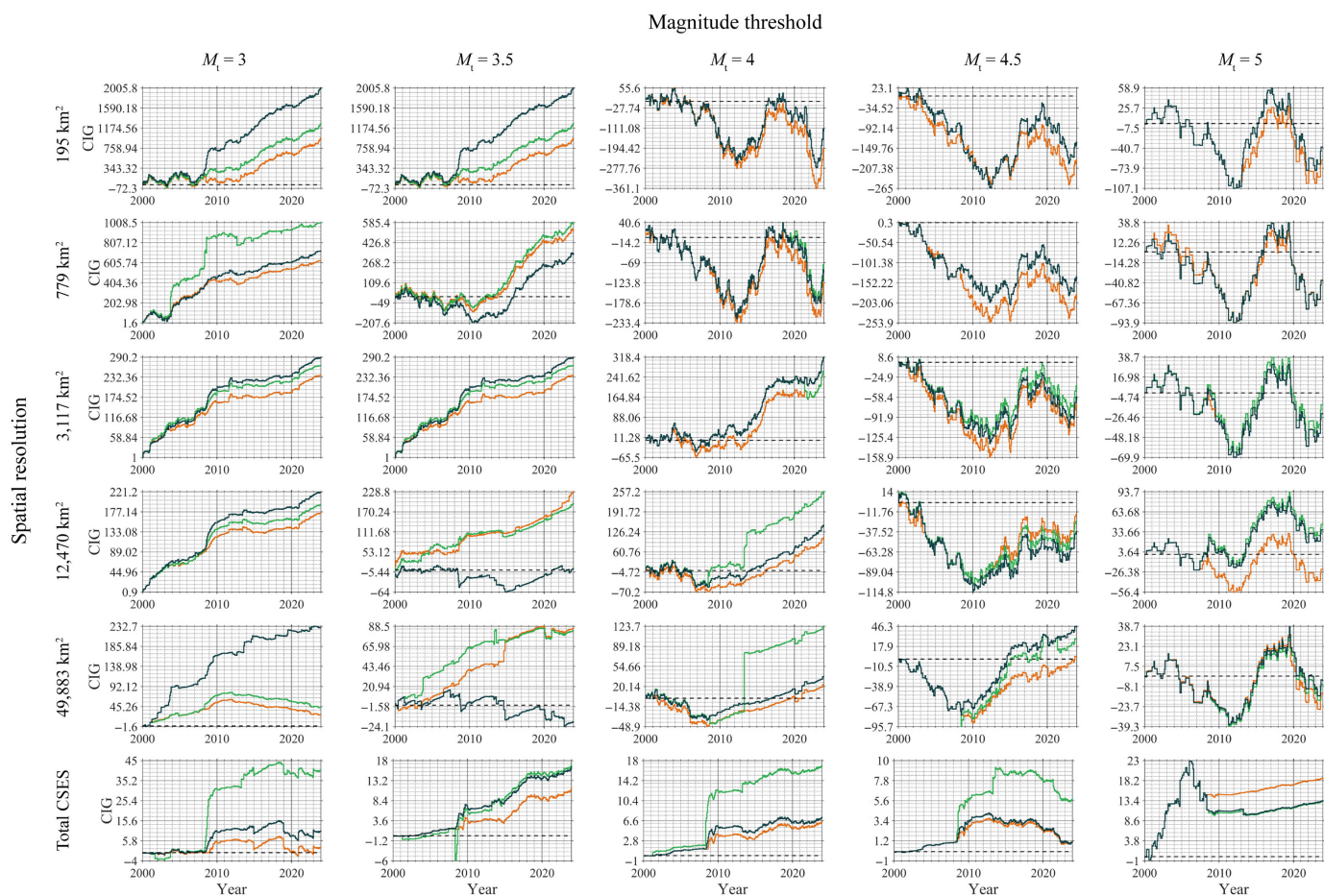


Figure 4. Time series of cumulative information gain (CIG) of $ETAS_{\mu(x,y)}$ over $ETAS_{\mu}$ in pseudopropective forecasting experiments at the CSES from 2000 to 2023. Results are shown for six spatial resolutions, five magnitude thresholds (M_t), and three strategies for assigning $P(n_i)$ when the observed value n_i in the i th triangle exceeds the forecasted maximum. The green, black,

and orange curves correspond to strategies (1), (2), and (3), respectively, described in Figure 1. The pseudopropective forecasting experiments use calendar months as time windows, resulting in a total of 288 time windows over the study period. The horizontal dashed black line indicates zero CIG. The color version of this figure is available only in the electronic edition.

TABLE 1

Percentage of Out-of-Range Grid Cells Across All 288 Time Windows for Each Combination of Spatial Resolution and Magnitude Threshold M_t

Spatial Resolution	$M_t=3$	$M_t=3.5$	$M_t=4$	$M_t=4.5$	$M_t=5$
195 km ²	0.36%	0.51%	0.73%	1.36%	1.74%
779 km ²	0.15%	0.22%	0.40%	0.52%	49.78%
3,117 km ²	0.06%	0.12%	0.16%	26.07%	43.96%
12,470 km ²	0.03%	0.05%	11.19%	14.82%	40.12%
49,883 km ²	0.01%	0.38%	5.39%	7.41%	41.11%
Total CSES region	1.21%	1.69%	2.44%	4.43%	26.11%

Here, an out-of-range grid cell refers to a case in which the maximum earthquake count obtained from the simulated catalogs is smaller than the corresponding observed earthquake count. The percentage is calculated relative to the total number of grid cells in each configuration. CSES, China Seismic Experimental Site.

probability—that is, when the observed earthquake count exceeds the maximum simulated count of both models—their LL scores become identical, resulting in an IG of zero. In contrast, when the observed count exceeds the maximum simulated value of only one model but not the other, the former adopts the fixed minimum probability, and the latter still uses the adaptive Gaussian kernel density estimate to assign $P(n_i)$. This asymmetric treatment introduces a strong bias in model performance. Such cases frequently occur in the months following large earthquakes (e.g., events with $M > 7$); examples are observed for $M_t = 3.0$ during June and November 2008 in the full CSES region (see [Data and Resources](#)).

During these postmainshock forecast windows, $\text{ETAS}_{\mu(x,y)}$ systematically generates more tail events in the probability distribution $P(n)$ than ETAS_{μ} . This difference arises because the spatially varying $\text{ETAS}_{\mu(x,y)}$ incorporates a heterogeneous background rate $\mu(x,y)$, which accounts for spatial clustering and variability in triggering intensity. Statistically, this produces a mixture of local Poisson-branching processes with different means, naturally leading to a fatter-tailed distribution. Physically, high- μ regions act as persistent triggering hotspots, sustaining multiple generations of aftershocks and resulting in abnormally large event counts within certain cells. Consequently, $\text{ETAS}_{\mu(x,y)}$ exhibits a broader and more asymmetric $P(n_i)$, consistent with the enhanced spatial heterogeneity of real seismicity.

In fact, the exclusion strategy for assigning $P(n_i)$ suffers from a similar problem, though its impact on CIG is less pronounced than that of the fixed minimum probability. Specifically, when both models assign probabilities under the exclusion rule (i.e., when the observed count exceeds the simulated maximum of both models), the IG of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ} is zero. When the observed count exceeds the simulated range of only one model, the LL for that model cannot be computed, and thus the IG for this forecast window becomes undefined. In practice, this window is omitted from the IG calculation, which explains why its impact on the overall CIG is less pronounced than that of the fixed minimum probability approach. In either case, whether using the fixed minimum probability or exclusion, the model performance is systematically biased following large earthquakes. Therefore, neither method should be recommended for evaluating IG in such contexts.

Spatial patterns of forecast performance

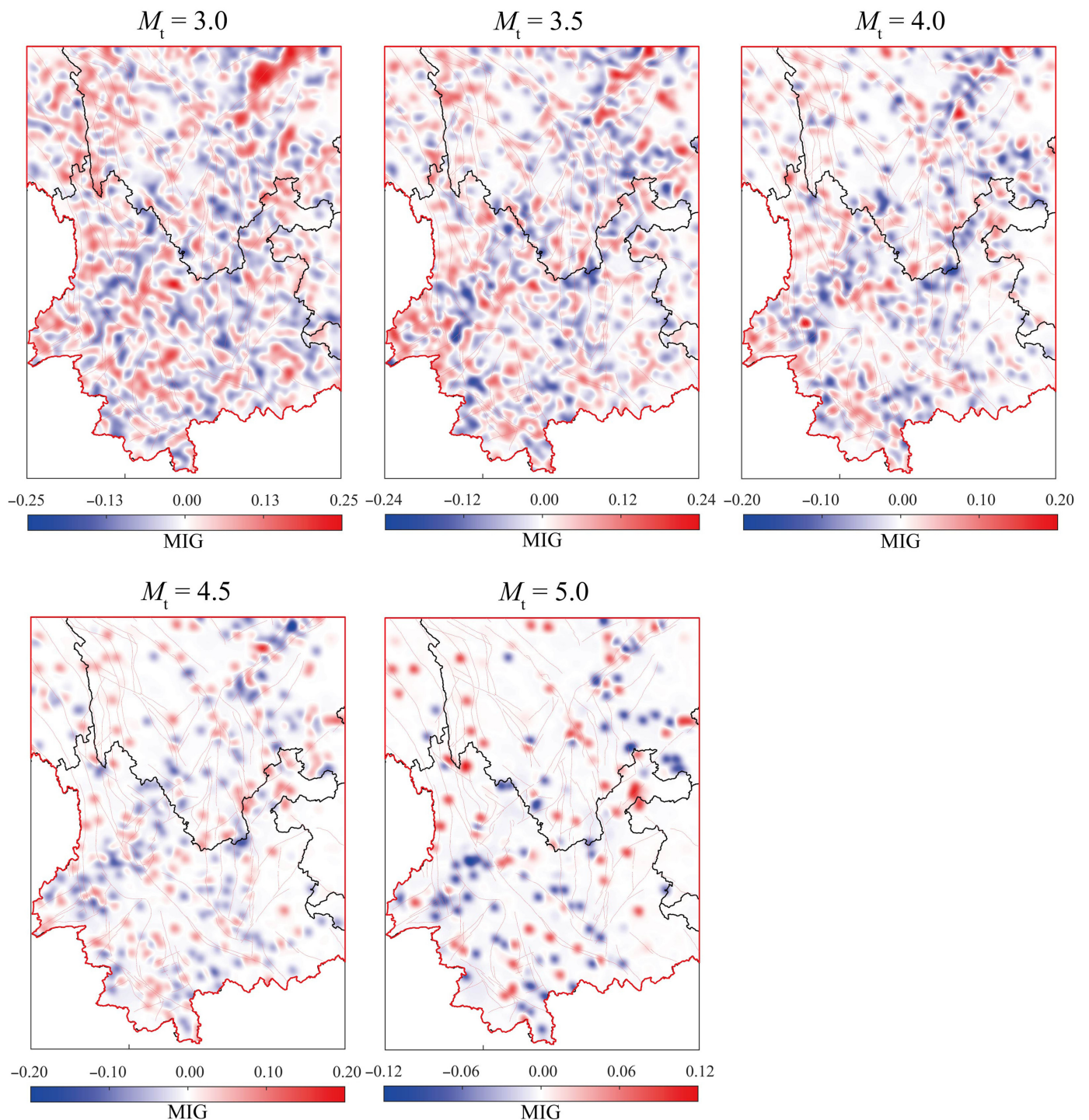
To further examine the spatial dependence of forecast performance, we mapped the MIG of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ} across all 288 monthly forecast windows from 2000 to 2023 (Fig. 5). The analysis was performed at the finest spatial resolution considered in the present study, corresponding to triangular cells of $\sim 195 \text{ km}^2$. For each cell, the IG was first calculated for each forecast window as the LL difference between $\text{ETAS}_{\mu(x,y)}$ and the benchmark ETAS_{μ} model and then averaged over the full study period. In calculating the IG, out-of-range observations were handled using the power law extrapolation strategy,

which has been shown in the [CIG of \$\text{ETAS}_{\mu\(x,y\)}\$ over \$\text{ETAS}_{\mu}\$](#) and [Bias from improper tail treatment](#) sections to provide the most theoretically justified and empirically robust treatment of the tail of the forecasted probability distribution $P(n)$. Therefore, Figure 5 provides a spatially resolved summary of the relative forecasting skill of $\text{ETAS}_{\mu(x,y)}$ compared with ETAS_{μ} after accounting for the fat-tailed nature of earthquake-count distributions, which is discussed in detail in the [Consistency of empirical scaling and ETAS-based theoretical predictions](#) section.

The resulting map (and Fig. 2) shows that the regions where $\text{ETAS}_{\mu(x,y)}$ achieves positive MIG are not randomly distributed. Instead, they are spatially concentrated in areas that experienced earthquakes with $M \geq 5$ during the study period. These positive-gain regions indicate locations where the spatially heterogeneous background model provides a better description of the observed earthquake-count distribution than the homogeneous benchmark model. In contrast, regions without such moderate to large earthquakes generally exhibit weaker or more neutral differences between the two models. This spatial correspondence suggests that the advantage of $\text{ETAS}_{\mu(x,y)}$ is most evident in areas where seismicity is sufficiently active and clustered to provide informative constraints on the spatial variability of background seismicity and triggering behavior.

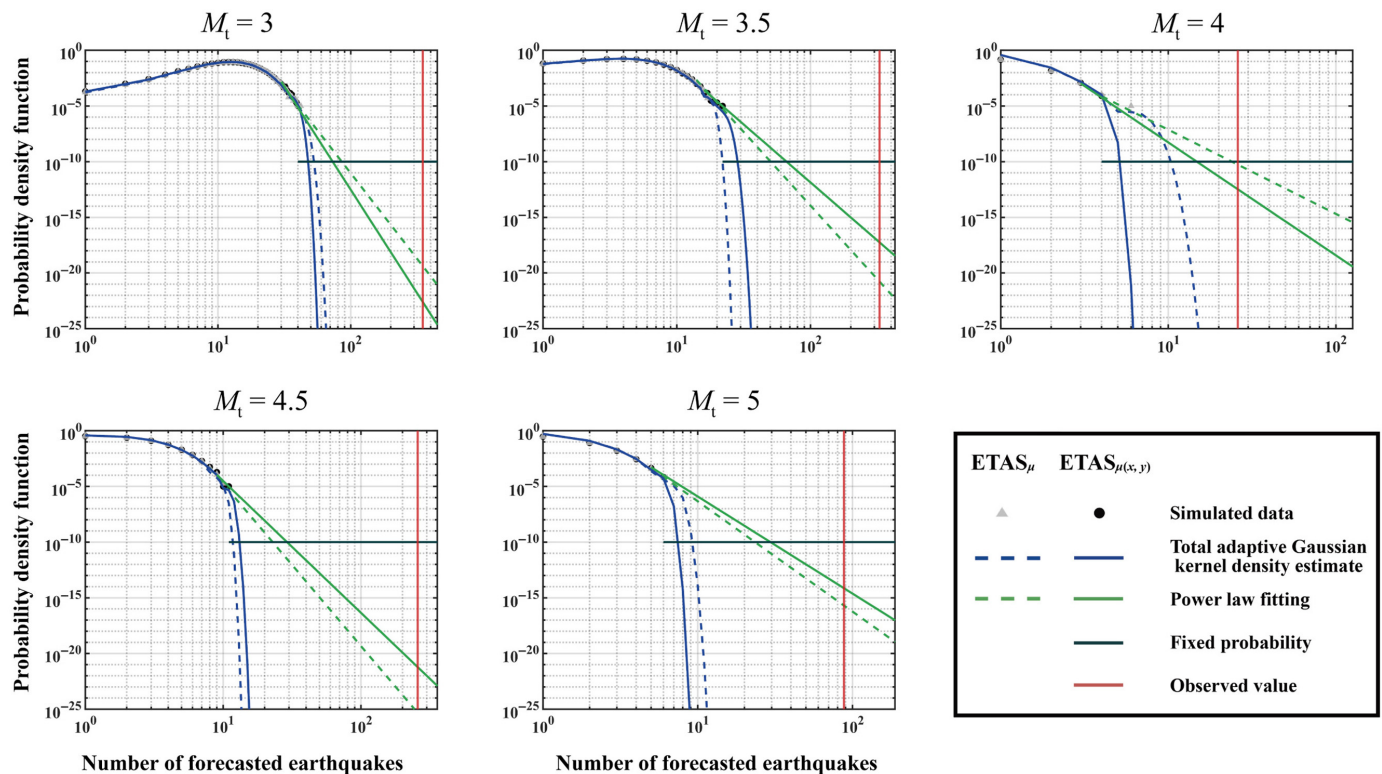
This pattern can be understood from both statistical and physical perspectives. Statistically, the spatially variable background rate $\mu(x,y)$ allows $\text{ETAS}_{\mu(x,y)}$ to redistribute long-term seismicity rates according to the observed spatial heterogeneity of the catalog, whereas ETAS_{μ} imposes a spatially uniform background rate over the entire region. As a result, in cells located near repeatedly active fault zones or near areas affected by $M \geq 5$ earthquakes and their aftershock sequences, $\text{ETAS}_{\mu(x,y)}$ can assign more realistic probabilities to elevated earthquake counts. Physically, these regions correspond to persistent or recurrent seismogenic structures where background activity, local triggering, and aftershock cascades are strongly localized in space. The heterogeneous background field therefore acts as a mechanism for incorporating such spatially persistent seismicity, improving the forecasted PDFs $P(n)$ in active regions. By contrast, in relatively quiet areas, the number of observed events is small, and the forecast evaluation is more strongly affected by stochastic fluctuations, making the performance difference between the two models less systematic.

The 2008 Wenchuan sequence also raises the question of whether a single extreme cluster can disproportionately control the overall model ranking. We note that the present study does not rely solely on a single terminal ~ 25 -yr score but presents the forecast evaluation as CIG time series in Figure 4. This representation makes the contribution of large earthquake sequences explicit and temporally localized: the influence of the Wenchuan sequence can be identified from the abrupt change in the CIG curves around 2008, whereas the CIG increments before and after this period remain separately interpretable. Moreover, large



earthquake sequences should not be regarded simply as outliers to be removed from forecast evaluation. They are precisely the periods when short-term earthquake forecasting is most important, and the ability of a model to assign reasonable probabilities to aftershock-rich windows is an essential component of forecasting skill. Excluding the most extreme windows would therefore provide a conditional evaluation of model performance during relatively quiet periods but would no longer assess the full pseudoprospective forecasting problem considered here. The spatial pattern of MIG in Figure 5 further shows that

Figure 5. Mean information gain (MIG) of $\text{ETAS}_{\mu(x,y)}$ over ETAS_{μ} across a total of 288 time windows during the study period in pseudoprospective forecasting experiments for the CSES from 2000 to 2023 at a spatial resolution of 195 km^2 . The information gain used to compute the MIG is based on the power law extrapolation method. The red colors in the color bar indicate positive MIG, meaning that $\text{ETAS}_{\mu(x,y)}$ outperforms ETAS_{μ} . The color version of this figure is available only in the electronic edition.



the positive performance of $ETAS_{\mu(x,y)}$ is concentrated mainly in regions where $M \geq 5$ earthquakes occurred during the study period, indicating that the gain reflects physically meaningful clustered seismicity rather than an arbitrary artifact of a single extreme observation.

Discussion

In this section, we present the results of pseudoprospective experiments conducted for May 2008 across the entire CSES region to quantitatively examine the impact of out-of-range earthquake counts on the calculation of IG. Following this, we further explain the underlying causes of noisy CIG, highlighting both methodological and data-related factors. Finally, we examine the empirical scaling behavior of earthquake-count distributions and discuss its consistency with ETAS-based branching-process theory, including its implications for constraining the fertility exponent α .

An example of out-of-range earthquake counts: The May 2008 Wenchuan earthquake

Within the spatiotemporal window of the present study, the largest event is the M_s 8.0 Wenchuan earthquake of 12 May 2008. This mainshock and its aftershocks produced an out-of-range situation, as illustrated in Figure 1b, under all magnitude thresholds (M_t) considered in the present analysis (see Fig. 6). Figure 6 also presents the probability distributions of earthquake counts forecasted by $ETAS_{\mu}$ and $ETAS_{\mu(x,y)}$ for the same month (May 2008) across the entire CSES region. The results clearly demonstrate that when the observed counts fall outside the simulated ensemble, the extrapolation based on

Figure 6. Probability distributions of earthquake counts forecasted by (dashed line) $ETAS_{\mu}$ and (solid line) $ETAS_{\mu(x,y)}$ for May 2008 across the entire CSES region, under different magnitude thresholds (M_t). The M_s 8.0 Wenchuan earthquake on 12 May and its aftershocks (marked by the red vertical line) generate an out-of-range situation as shown in Figure 1b, which corresponds to a single time-window slice taken from the last row of Figure 4. All plotting elements are consistent with those in Figure 1, except that the $ETAS_{\mu}$ data points are shown as solid gray triangles for visual distinction. The [Data and Resources](#) section provides links to access the forecasted probability density functions of both models for all forecast windows and magnitude thresholds across the full CSES region. The color version of this figure is available only in the electronic edition.

adaptive Gaussian KDE becomes severely biased and drastically underestimates the probability of the observed earthquake numbers. In contrast, power law distributions with fat-tailed behavior provide a more reliable extrapolation, yielding non-negligible probabilities for these extreme counts.

From Figure 6, the IG of $ETAS_{\mu(x,y)}$ relative to $ETAS_{\mu}$ can be computed using equation (3). The corresponding IG values are $\sim 2, -2, -1, -3$, and 1 for $M_t = 3, 3.5, 4, 4.5$, and 5 , respectively. These results suggest that at the time of the Wenchuan mainshock itself, the two models did not exhibit a consistently large performance gap across thresholds. However, the bottom row of Figure 4 reveals a critical point: when extrapolation is performed using power law tails, the IG of $ETAS_{\mu(x,y)}$ relative to $ETAS_{\mu}$ continues to rise in the months following May 2008. This indicates that $ETAS_{\mu(x,y)}$ systematically outperforms $ETAS_{\mu}$ in forecasting the Wenchuan aftershocks, leading to the marked upward shift observed in the CIG curve in Figure 4.

TABLE 2

Sensitivity of Power Law Extrapolation to the Upper-Tail Fraction Used for Power Law Fitting

Tail fraction	$M_t=3$		$M_t=3.5$		$M_t=4$		$M_t=4.5$		$M_t=5$	
	ETAS _{μ}	ETAS _{$\mu(x,y)$}	ETAS _{μ}	ETAS _{$\mu(x,y)$}	ETAS _{μ}	ETAS _{$\mu(x,y)$}	ETAS _{μ}	ETAS _{$\mu(x,y)$}	ETAS _{μ}	ETAS _{$\mu(x,y)$}
10%	-15.49	-23.43	—	-13.79	—	—	—	—	—	—
20%	-18.04	-18.43	-21.87	-10.72	-29.21	—	—	—	—	—
30%	-19.37	-17.28	-19.73	-21.34	-22.88	-23.46	-14.69	-17.49	-9.28	-13.03
40%	-16.07	-15.30	-16.45	-19.67	-22.38	-20.38	-15.63	-15.56	-11.07	-12.81
50%	-12.30	-12.10	-13.74	-16.23	-15.97	-18.19	-16.31	-13.22	-11.07	-10.32

Values are $\log_{10} P(n_i)$, in which $P(n_i)$ is the forecasted probability assigned to the observed earthquake count n_i by ETAS _{μ} and ETAS _{$\mu(x,y)$} for May 2008 across the entire China Seismic Experimental Site (CSES) region, under different magnitude thresholds M_t . The em dash indicates cases in which no stable extrapolated probability was obtained. ETAS, epidemic-type aftershock sequence.

To examine whether the choice of the upper 30% tail fraction introduces a substantial bias in the power law extrapolation, we performed a sensitivity test by repeating the extrapolation using upper-tail fractions from 10% to 50% of the simulated data (Table 2). This test is intended to assess the stability of the extrapolated likelihood for the May 2008 out-of-range case rather than to infer the theoretical scaling exponent a discussed in the [Consistency of empirical scaling and ETAS-based theoretical predictions](#) section. The results show that the 10% and 20% choices do not always provide stable estimates across all magnitude thresholds and both ETAS variants, as indicated by missing entries in several cases. This suggests that using too small a fraction of the simulated tail may leave insufficient support for a robust extrapolation, especially when the forecasted distribution becomes sparse at higher magnitude thresholds. In contrast, tail fractions of 30% to 50% provide finite estimates for all magnitude threshold values M_t and for both ETAS _{μ} and ETAS _{$\mu(x,y)$} . Although the resulting $\log_{10} P(n_i)$ values vary with the chosen tail fraction, as expected for far-tail extrapolation, they consistently indicate extremely small but nonzero probabilities for the May 2008 out-of-range observations. The qualitative conclusion therefore remains unchanged: the Wenchuan mainshock and its aftershocks generated earthquake counts far beyond the simulated range, and power law extrapolation provides a more appropriate treatment of such extreme observations than either assigning a fixed minimum probability or excluding the corresponding likelihood term. We therefore adopted the upper 30% tail fraction as the default choice because it is the smallest fraction that yields complete estimates across all tested configurations while still focusing on the upper part of the simulated distribution. This choice represents a practical bias–variance compromise and does not drive the main conclusions of the present study.

Quantitative rationale for the variability of CIG

Figure 4 shows that the CIG curves of ETAS _{$\mu(x,y)$} relative to ETAS _{μ} lose their long-term upward trend and become

increasingly erratic when the spatial resolution is coarsened and M_t is raised. Subsequently, we offer a quantitative rationale for this behavior.

Let n_{ik} be the number of earthquakes observed in the k th grid cell at the chosen resolution r during the i th forecast window, for which there are N_r grid cells, with $k = 1, \dots, N_r$. The number of grid cells scales as $N_r \sim 1/r$. Let us assume that n_{ik} follows a true discrete probability law $P_0(n)$ with mean $n_0 > 0$. Suppose models A and B issue distinct forecast distributions $P_{A(n)}$ and $P_{B(n)}$ with means n_A and n_B , respectively. Although the condition $|n_A - n_0| < |n_B - n_0|$ indicates that model A provides a more accurate mean forecast, it is not sufficient by itself to establish superior probabilistic performance. We therefore assume that model A is better than model B in the expected log-score sense: $E_{P_0}[\log P_{A(N)} - \log P_{B(N)}] > 0$. Define the per-cell LL difference for the k th grid cell during the i th forecast window:

$$\Delta LL_{ik} = \log P_{A,ik}(n_{ik}) - \log P_{B,ik}(n_{ik}). \quad (6)$$

Under the expected log-score assumption above, the expected LL difference is positive, indicating that model A provides the better probabilistic fit. The relative variability of this statistic is assessed using the coefficient of variation (CV) of the LL difference:

$$CV_i = \frac{\text{Std}_k(\Delta LL_{ik})}{|\text{Mean}_k(\Delta LL_{ik})|}, \quad (7)$$

Here, the mean and standard deviation are calculated over the set of N_r cells, $k = 1, \dots, N_r$, separately for each forecast window i . The absolute value ensures that CV_i is nonnegative; however, CV_i becomes unstable when $\text{Mean}_k(\Delta LL_{ik})$ is close to zero and should therefore be interpreted cautiously in such cases. The CV is a function of the resolution r through the dependence of the number of grid cells $N_r \sim 1/r$. The coarser the resolution r is, the larger is n_0 and the narrower is $P_0(n)$, but the smaller is N_r . Thus coarsening the resolution r improves the statistics per cell but deteriorates the statistical accuracy obtained by averaging

TABLE 3

Coefficient of Variation (CV) Calculated Based on the Cumulative Information Gain (CIG) Results for ETAS _{$\mu(x,y)$} Relative to ETAS _{μ} in the China Seismic Experimental Site (CSES) Region, Evaluated for Each Combination of Spatial Resolution and Magnitude Threshold M_t

Spatial Resolution (km ²)	$M_t=3$	$M_t=3.5$	$M_t=4$	$M_t=4.5$	$M_t=5$
195	7.28	15.73	22.92	16.02	34.62
779	3.87	7.19	28.18	13.38	48.38
3,117	3.08	4.64	9.05	27.25	68.67
12,470	2.62	3.42	13.03	93.82	54.27
49,883	10.90	4.72	25.78	21.50	64.33

The CV values are calculated from the exclusion strategy only, that is, from the orange CIG curves shown in Figure 4.

over all N_r cells. This competition between in-cell and ensemble-cell statistics explains the complicated dependences of the CIG curves on the resolution and magnitude threshold shown in Figure 4.

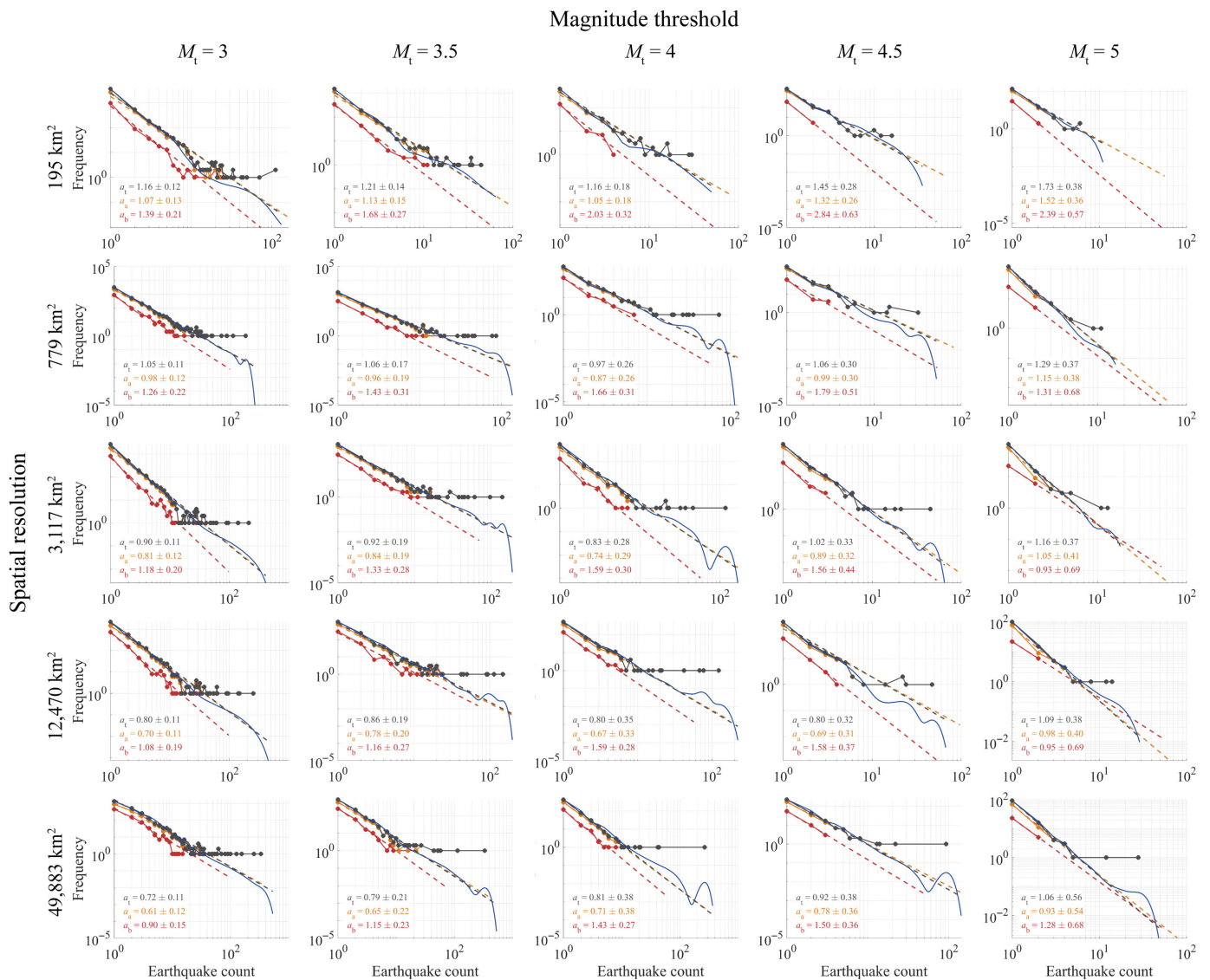
Table 3 reports the CV computed solely from the orange CIG curves in Figure 4, that is, using the exclusion strategy (see more details in the [Distribution of forecasted earthquake numbers](#) section). Under this strategy, only the probabilities obtained from the adaptive-Gaussian kernels are retained; any cells for which observed counts fall outside the simulated range are discarded. This ensures that all CV values are derived from a single, internally consistent method, without the complications introduced by fixing minimum probability or power law extrapolation. The results show a clear pattern: for a given spatial resolution, a larger M_t lowers the average count n_0 (e.g., Figs. 6 and 7); the mean of ΔLL therefore declines while its sampling variance grows, and the CV increases accordingly.

The CV values in Table 3 should be interpreted as an empirical signal-to-noise diagnostic for the stability of the CIG curves rather than as a formal hypothesis test. In general, configurations with small CV values (<10) show relatively stable CIG behavior and are therefore more robustly interpretable. Configurations with intermediate CV values, between ~ 10 and 20, should be interpreted together with the corresponding CIG time series in Figure 4 because some of them still exhibit coherent long-term trends despite moderate variability. For example, several low-threshold configurations with $M_t = 3$ or $M_t = 3.5$ display persistent positive CIG trends even when their CV values are slightly larger than 10. In contrast, configurations with CV values well above 20 are increasingly dominated by stochastic variability, and extreme cases, such as CV = 93.82 for $M_t = 4.5$ at 12,470 km², should not be interpreted as providing reliable evidence for systematic model superiority or inferiority. Overall, Table 3 and Figure 4 jointly indicate that statistically interpretable CIG results are mainly obtained for data-rich settings, especially lower magnitude thresholds and configurations with persistent CIG trends, whereas high- M_t and sparse-count settings are largely noise dominated.

Consistency of empirical scaling and ETAS-based theoretical predictions

Figure 7 demonstrates that the empirical frequency distributions of earthquake counts per grid cell are well approximated by the power law $P(n) \sim n^{-(1+a)}$ across all tested magnitude thresholds and spatial resolutions. In Figure 7, the scaling behavior is further examined separately for the full study period, the pre-Wenchuan period, and the post-Wenchuan period to evaluate whether the 2008 M_s 8.0 Wenchuan earthquake produces a systematic change in the fitted exponent. For each combination of spatial resolution and magnitude threshold, the mean and standard deviation of the fitted exponent a were estimated from 200 bootstrap resamples. Figure 8 then plots the histogram of the 25 mean a values shown in Figure 7 together with their Gaussian fit for the full study period, the period before Wenchuan, and the period after Wenchuan. The corresponding ensemble-averaged exponents are $a_t = 1.02 \pm 0.23$, $a_b = 1.47 \pm 0.45$, and $a_a = 0.92 \pm 0.22$, respectively. We therefore regard $a_t = 1.02 \pm 0.23$ as an effective ensemble-averaged exponent across spatial resolutions and magnitude thresholds rather than as an exact universal exponent for every individual configuration.

Using the technique of generating probability functions applied to the ETAS model, [Saichev et al. \(2005\)](#) derived the theoretical distribution of the total number of aftershocks, which follows the power law $P(n) \sim n^{-(1+a)}$ in the noncritical regime (i.e., when its branching ratio h is not close to 1 and for earthquake counts $n > 1/(1-h)^{g/(g-1)}$, in which $g = b/\alpha$, b is the Gutenberg–Richter (GR) slope and α is the fertility (or productivity) exponent describing how the mean number of aftershocks triggered by a mainshock scales with its magnitude. The power law exponent a is related to the model parameters through $a = g = b/\alpha$. For smaller event counts $n < 1/(1-h)^{g/(g-1)}$, the exponent a is renormalized to $a = 1/g = \alpha/b$. The empirical exponents obtained here are mostly close to or larger than unity within uncertainty, suggesting broad consistency with the branching-process scaling regime, although they should not be interpreted as a strict



diagnostic of criticality in finite and nonstationary observational windows. This finding corroborates previous studies suggesting that the Earth's crust, on average, remains far from criticality (Nandan, Ram, *et al.*, 2021; Li, Sornette, Wu, *et al.*, 2025). Saichev and Sornette (2006) further demonstrated that this same scaling law emerges as the asymptotic behavior of space-time windowed event-number distributions, implying that such power laws are a fundamental statistical property of branching-type seismicity models.

We emphasize, however, that the theoretical result of Saichev *et al.* (2005) strictly applies to the total progeny of an idealized stationary branching process in unbounded space and time. In contrast, the empirical distributions analyzed here are constructed from earthquake counts within finite one-month forecast windows, finite spatial cells, and fixed magnitude thresholds, in a region that exhibits temporal nonstationarity, particularly around the 2008 Wenchuan earthquake. Therefore, the comparison between theory and data should be interpreted in an approximate asymptotic sense rather than

Figure 7. Frequency distribution function of earthquake count per grid cell over the ensemble of grid cells, aggregated over all 288 forecast windows for the CSES. Each point represents the number of grid cells (on the y axis) that experienced a specific number of earthquakes (on the x axis) for a specific spatial resolution and magnitude threshold. In other words, the curves show the empirical nonnormalized probability density function of earthquake counts per cell. The black, orange, and red fitted lines denote the power law fits for the full study period, the pre-Wenchuan period, and the post-Wenchuan period, respectively. The corresponding exponents are denoted by a_t , a_b , and a_a . Consistent with Figure 1, the solid blue line represents the adaptive Gaussian kernel density estimate, and the dashed lines show the fitted power law function ($P(n) \sim n^{-(1+a)}$). For each combination of spatial resolution and magnitude threshold, the mean and standard deviation of the fitted exponent a were estimated from 200 bootstrap resamples. The color version of this figure is available only in the electronic edition.

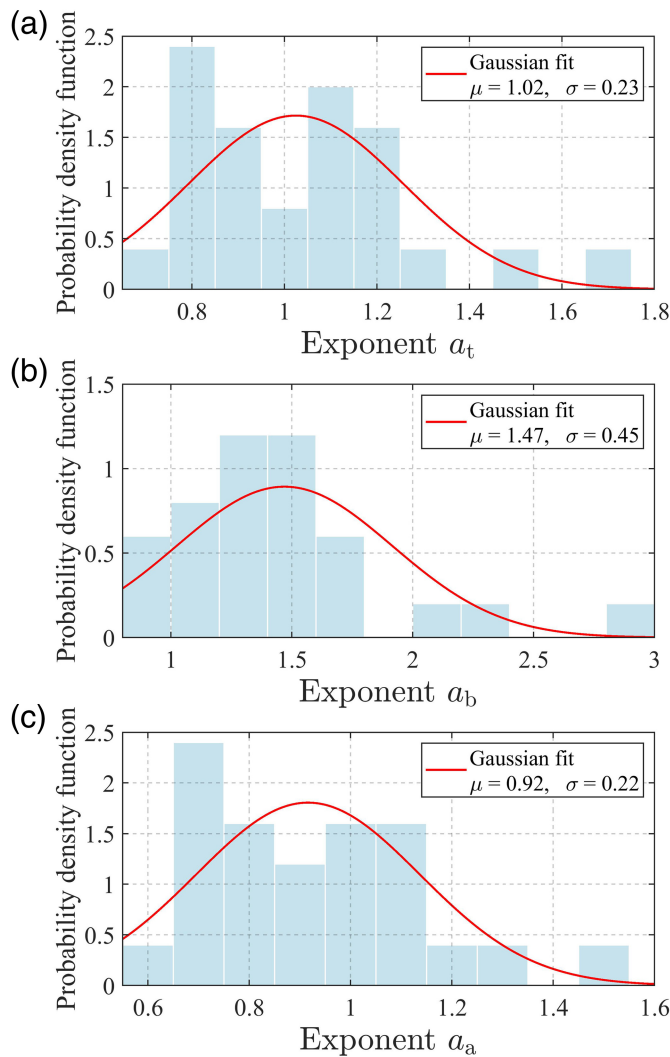


Figure 8. Histograms of the 25 mean power law exponents a shown in Figure 7 for the (a) total, (b) pre-, and (c) post-Wenchuan 2008 earthquake and their Gaussian fit. The color version of this figure is available only in the electronic edition.

as a strict finite-sample validation of the idealized theory. The relevance of the theory in the present setting is supported mainly by the extension of [Saichev and Sornette \(2006, 2007\)](#), who showed that similar power law scaling can emerge for event-number distributions measured in finite space-time windows. Under realistic observational constraints, finite temporal windows, spatial truncation, boundary effects, and magnitude thresholds are expected to distort the distribution, especially at small earthquake counts. Accordingly, the theoretical scaling is expected to be most meaningful in the intermediate- to large-count regime, which is the part of the distribution used for tail fitting in the present study.

Figure 7 also provides a direct check on the possible influence of the 2008 Wenchuan earthquake. Although local differences between the pre- and post-Wenchuan fitted exponents are visible in some panels, the two sets of estimates remain broadly

overlapping within their uncertainties determined by bootstrap. The pre-Wenchuan distribution yields a larger ensemble-averaged exponent, $a_b = 1.47 \pm 0.45$, whereas the post-Wenchuan distribution gives $a_a = 0.92 \pm 0.22$. This contrast points to a possible modification of the effective scaling behavior by the Wenchuan sequence and its aftermath; however, the overlap between the uncertainty ranges suggests that the two exponents may be statistically indistinguishable at this level of precision. Thus, the hypothesis that the pre- and post-Wenchuan distributions share a common scaling exponent cannot be rejected from these estimates alone. Instead, the Wenchuan earthquake represents a strong transient episode superimposed on a broader branching process-like scaling behavior. We also note that temporal variations in magnitude completeness, especially immediately after strong earthquakes, may affect the observed event counts. However, the persistence of similar scaling behavior across multiple magnitude thresholds, together with the catalog-completeness analysis discussed in the [Seismicity in CSES](#) section, suggests that short-term catalog incompleteness does not dominate the inferred scaling.

A systematic dependence of the fitted exponent a on the magnitude threshold M_t is nevertheless visible in Figure 7: lower magnitude thresholds tend to produce smaller fitted exponents, whereas higher magnitude thresholds tend to yield larger values. We interpret this trend primarily as an effective finite-sample and thresholding effect rather than as evidence against the branching-process scaling relation. Increasing M_t reduces the number of events available in each space-time bin and shifts the fitted distribution toward sparser and more discrete count statistics. At the same time, higher thresholds preferentially retain larger events and their more localized aftershock productivity so that the fitted tail can reflect a threshold-dependent effective exponent. Conversely, at lower M_t , the larger number of small events provides better sampling of the full cascade and tends to broaden the count distribution, leading to smaller fitted a . Thus, whereas the full-period value $a_t = 1.02 \pm 0.23$ should be viewed as a compact summary of the ensemble behavior across all tested configurations, the threshold-dependent deviations reflect finite-window sampling, magnitude-threshold effects, and residual nonstationarity in the observed catalog.

Moreover, the observed power law exponent a provides an indirect way to infer the ETAS fertility (or productivity) exponent α . Given the theoretical relation $a = g := b/\alpha$, the empirical result $a_t = 1.02 \pm 0.23$ and the Gutenberg–Richter slope value $b = 0.85 \pm 0.002$ (see [Li, Sornette, Wu, et al., 2025](#)) yield an estimated fertility exponent of $\alpha_t = b/a_t = 0.83 \pm 0.19$ for the full study period. Applying the same error-propagation calculation separately to the pre- and post-Wenchuan exponents gives $\alpha_b = 0.58 \pm 0.18$ and $\alpha_a = 0.92 \pm 0.22$, respectively. Here, the uncertainty of α was calculated by standard error propagation, assuming that the uncertainties in a and b are independent. Although the uncertainty in b is relatively small

compared with the variability in a , including it provides a more realistic estimate, especially given the mixed magnitude types in the CSES catalog. This inferred value of α_t falls well within the physically plausible range reported by previous ETAS-based studies and reinforces the interpretation that seismic triggering efficiency increases moderately with magnitude but remains close to the marginal regime ($\alpha \approx b$). The parameter estimates obtained directly from the ETAS-based models give (Fig. 3): $\alpha \approx 0.7\text{--}1.0$ for $\text{ETAS}_{\mu(x,y)}$ and $\alpha \approx 0.6\text{--}0.8$ for ETAS_{μ} , in good agreement.

Although the indirect estimate inferred from the count-distribution scaling is in good agreement with the directly calibrated ETAS parameters, the two estimates are not expected to be identical. Several factors may account for the remaining differences between them. First, whereas the theoretical derivation assumes an idealized, stationary branching process in the noncritical regime, the ETAS fitting is affected by nonstationary background rates, spatial heterogeneity, and catalog incompleteness, all of which can modify the apparent fertility. In particular, undetected small earthquakes can break the true parent–offspring structure of the cascade. Their descendants may then be attributed to larger observed events, making large earthquakes appear more productive than they actually are and thereby increasing the estimated α . This mechanism is closely related to the apparent-clustering bias caused by incomplete earthquake catalogs (Sornette and Werner, 2005b) and to the sensitivity of ETAS parameter estimation to short-term aftershock incompleteness (Omi *et al.*, 2014; Zhuang *et al.*, 2017; Mizrahi *et al.*, 2021; Hainzl, 2022; Hainzl *et al.*, 2024). Second, the estimation of α in ETAS models is strongly coupled with that of other parameters, particularly the background rate μ and the spatiotemporal decay parameters ($1 + \rho$ and $1 + \omega$), leading to potential trade-offs that shift the fitted values upward. For example, an underestimated background rate or an overly concentrated spatial triggering kernel can be compensated by a larger productivity exponent because the likelihood can improve by assigning more offspring to larger events. Third, the boundary and finite-size effects inherent in the CSES region (Li, Sornette, Wu, *et al.*, 2025), especially the presence of clustered aftershock sequences such as for the 2008 Wenchuan earthquake, may enhance the observed triggering efficiency relative to the long-term theoretical expectation.

Despite these complications, the full-period estimate $\alpha_t = 1.02 \pm 0.23$ remains broadly consistent with both the directly calibrated ETAS values and the Gutenberg–Richter slope $b = 0.85 \pm 0.002$. This indicates that the system lies close to the transition between small event–dominated and large event–dominated triggering. Therefore, rather than claiming a strong ultraviolet-divergence regime (Sornette and Werner, 2005a), our results support a more cautious interpretation: small earthquakes remain collectively important because of their large numbers, but the inferred productivity exponent is close enough to b that finite-window effects, catalog

incompleteness, and large aftershock sequences can substantially affect the apparent balance between small and large earthquake triggering. Because direct empirical estimation of the fertility exponent α is notoriously difficult, often hindered by limited temporal completeness and strong parameter correlations, our indirect inference offers a robust and independent constraint. It provides empirical support for theoretical expectations derived from branching process theory and anchors a key ETAS parameter on a statistically grounded foundation directly linked to the observed scaling behavior of earthquake count distributions.

Conclusions

Advancing short-term earthquake forecasting has always been a core objective of the emerging CSES, led by China Earthquake Administration. A critical step in this agenda is the rigorous testing of scientific hypotheses on earthquake predictability. To this end, we have developed AoyuX, a bespoke short-term predictability platform tailored to the CSES region, which is currently in its initial implementation stage. AoyuX embeds the ETAS model, one of the few frameworks that simultaneously delivers time, space, magnitude, and probability information and that synthesizes the current state of seismological knowledge. By embedding two variants of the ETAS model ETAS_{μ} with a homogeneous background rate and $\text{ETAS}_{\mu(x,y)}$ with a spatially variable rate, we carried out ~ 25 yr of monthly pseudoprospective experiments (2000–2023).

The calibration phase reveals systematic contrasts between the two variants. $\text{ETAS}_{\mu(x,y)}$ consistently assigns more events to the background than ETAS_{μ} , reflecting the added flexibility of a nonparametric background field. Several parameter time series exhibit abrupt shifts that coincide with major regional earthquakes, indicating that large events can instantaneously reshape the triggering environment as well as be associated with transient data issues such as earthquake incompleteness. Forecast assessments, expressed through CIG, show a clear hierarchy under data-rich conditions: for the full CSES domain and for $M_t = 3.0$ to 3.5 , $\text{ETAS}_{\mu(x,y)}$ maintains a persistent positive CIG over ETAS_{μ} . The spatial distribution of MIG further shows that the advantage of $\text{ETAS}_{\mu(x,y)}$ is concentrated mainly in regions where $M \geq 5$ earthquakes occurred during the study period rather than being uniformly distributed over the entire CSES region. However, in sparse-count configurations caused by fine spatial discretization, high-magnitude thresholds, or limited seismicity in many cells, sampling variance grows, and the CIG curves become highly volatile. The CV analysis provides an empirical signal-to-noise diagnostic: whereas low-CV configurations yield more stable and interpretable CIG curves, very large CV values indicate that model comparison is dominated by stochastic variability rather than by coherent forecasting skill.

Model ranking also depends sensitively on how “out-of-range” cell counts, observed counts exceeding all simulated values, are handled. Among the strategies tested, power law

extrapolation provides the most theoretically justified and empirically robust results, in line with both analytical predictions of branching-process theory and the clear scaling patterns shown in Figure 7 and earlier studies (Saichev *et al.*, 2005; Saichev and Sornette, 2006; Saichev and Sornette, 2007). Although in a few coarse-grained low-magnitude configurations, the fixed minimum probability rule produced slightly higher CIG values, these are best interpreted as artifacts of noise rather than genuine model superiority. Both the fixed minimum probability and exclusion strategies introduce systematic biases in model evaluation following large earthquakes. Consequently, these approaches are not suitable for reliable assessment of IG during periods of intense seismic activity. Taken together, the evidence strongly supports power law extrapolation as the most appropriate framework for handling extreme counts for the ETAS-based branching models studied here, especially in cells affected by large, cluster-forming earthquakes, which generate the fat-tailed distributions expected from aftershock cascades. This highlights the necessity of treating such extreme outliers not as anomalies to be truncated but as integral signatures of the underlying triggering dynamics.

Although the present study focuses on ETAS-based forecasts, the proposed tail-aware scoring framework is not intrinsically tied to the ETAS model. Its essential requirement is that a forecasting model can provide, either analytically or through simulations, the full probability distribution $P(n)$ of earthquake counts in each space–time–magnitude bin. This condition can also be satisfied by machine learning–based forecasts, physics-based models, or hybrid approaches, provided that they generate probabilistic or ensemble forecasts rather than only deterministic point estimates. In such applications, however, the appropriate tail treatment should be diagnosed from the model-specific and data-specific count distributions. For ETAS-type branching models, power law extrapolation is theoretically motivated and empirically supported by the fat-tailed distributions observed here. Other model classes may produce different tail behavior (e.g., exponential, lognormal, or bounded distributions), in which case the extrapolation rule should be adapted accordingly. Therefore, the broader contribution of the present framework is not the exclusive use of a power law tail but the explicit recognition that out-of-range observations must be handled by a transparent, empirically testable, and physically interpretable tail model. This makes the framework potentially useful for fair model comparison across ETAS, machine learning, physics-based, and future hybrid earthquake forecasting models.

As a complementary result, our analysis provides independent estimates of the fertility exponent inferred from the empirical scaling of earthquake-count distributions and the ETAS-based branching-process theory. For the full study period, we obtain $a_t = 1.02 \pm 0.23$, which, together with $b = 0.85 \pm 0.002$, yields $\alpha_t = 0.83 \pm 0.19$. The pre- and post-Wenchuan periods yield $\alpha_b = 0.58 \pm 0.18$ and

$\alpha_a = 0.92 \pm 0.22$, respectively. These values should be interpreted as effective estimates under finite space–time windows and nonstationary catalog conditions rather than as exact universal constants. Nevertheless, they provide useful independent constraints on this key but difficult-to-estimate ETAS parameter and indicate that the productivity exponent is close to the Gutenberg–Richter b value in the CSES region.

In sum, $\text{ETAS}_{\mu(x,y)}$ provides demonstrable skill over its homogeneous counterpart in regimes where observational counts are plentiful and seismicity is sufficiently active and clustered. Where data are sparse, however, sampling noise, parameter instability, and treatment of extreme clusters become dominant concerns. Future work should therefore focus on adaptive strategies for rare-event handling, model-specific tail diagnostics, and on hybrid schemes that combine ETAS-style triggering with physics-based constraints, further enhancing the operational value of AoyuX for short-term earthquake forecasting in the CSES and analogous natural laboratories worldwide.

Data and Resources

The earthquake catalogs used in this study were obtained from the National Earthquake Science Data Center of China (https://data.earthquake.cn/datashare/report.shtml?PAGEID=earthquake_zhengshi, last accessed September 2026), which is publicly accessible; however, user registration is required to access the catalog information. The catalog of earthquakes with magnitudes ≥ 3.0 in the China Seismic Experimental Site (CSES) region from 1970 to 2023 is provided as Data Set S1 in Li, Sornette, Wu, *et al.* (2025). The simulated earthquake catalogs generated by the two epidemic-type aftershock sequence models can also be accessed at Li, Zhang, and Sornette (2025). In addition, the forward monthly, quarterly, semiannual, and annual AoyuX earthquake forecast reports for the CSES since June 2024 are available at Li (2025a). Links to the forecasted probability density functions of both models for all forecast windows and magnitude thresholds across the CSES region are available at Li (2025b,c,d,e,f).

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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